

Q1 Language of kinematics

Name: _____

Class: _____

Date: _____

Time: **282 minutes**

Marks: **233 marks**

Comments:

EXAM PAPERS PRACTICE

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Q1.

A particle moves in a straight line so that at time t seconds it has velocity v m s⁻¹, where

$$v = 3t^2 - 5t.$$

- (a) Find an expression for the acceleration of the particle at time t .

(2)

- (b) Find the value of t when the acceleration is zero.

(1)

(Total 3 marks)

Q2.

Fiona is cycling along a straight horizontal road when she spots her friend, Marie, ahead of her. Marie is jogging along the same road and in the same direction.

Fiona is cycling with constant speed 6 m s⁻¹ and Marie is jogging with constant speed 3 m s⁻¹.

- (a) The velocity-time graph below shows Fiona's motion from the moment when she spots Marie, when $t = 0$, until she reaches a bridge, when $t = T$.



Marie takes 2 seconds longer than Fiona to reach the bridge.

Copy the graph and draw another line to show Marie's motion from $t = 0$ until she reaches the bridge. Label the axes appropriately.

(2)

- (b) When $t = 0$, Marie is x metres from the bridge and Fiona is 20 metres behind her.

- (i) Write down, in terms of x , the distance travelled by Fiona.

(1)

- (ii) Write down an equation in terms of x and T for Fiona's motion.

(1)

- (iii) Write down an equation in terms of x and T for Marie's motion.

(2)

(iv) Hence find the values of x and T .

(4)

(Total 10 marks)

Q3.

A car is travelling along a straight horizontal road at a speed of 18 m s^{-1} when the driver sees a red traffic light ahead. The car then slows down uniformly, so that 6 seconds later the speed of the car is 9 m s^{-1} .



(a) While the car is slowing down:

(i) show that its acceleration is -1.5 m s^{-2} ;

(3)

(ii) find the distance that it travels.

(2)

(b) After the car has slowed down for 6 seconds, the traffic lights change to green. The car then accelerates at a constant 1.2 m s^{-2} and passes the traffic lights at a speed of 15 m s^{-1} .

(i) Show that the car accelerated for 5 seconds before reaching the traffic lights.

(2)

(ii) Find the distance of the car from the traffic lights when the car started to accelerate.

(3)

(c) Find the distance of the car from the traffic lights when the car began to slow down.

(1)

(Total 11 marks)

Q4.

A particle moves so that, at time t seconds, its position vector, $\mathbf{r}$ metres, is given by

$$\mathbf{r} = 4e^{-t}\mathbf{i} + (6t + 3e^{-t})\mathbf{j}$$

The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are perpendicular.

(a) Show that the velocity of the particle is $(-4\mathbf{i} + 3\mathbf{j}) \text{ m s}^{-1}$ when $t = 0$.

(3)

- (b) Find an expression for the acceleration of the particle at time t . (2)
 - (c) Find the magnitude of the acceleration of the particle when $t = 0$. (2)
 - (d) Describe what happens to the velocity of the particle as t becomes large. (2)
- (Total 9 marks)

Q5.

A car is travelling along a straight horizontal road. When the car is at a distance of 48 metres from a set of traffic lights, it is travelling at 20 m s^{-1} . The car slows down uniformly so that it has a speed of 4 m s^{-1} as it passes the traffic lights.

- (a) Show that the acceleration of the car is -4 m s^{-2} . (2)
- (b) How far past the traffic lights does the car travel before it stops? (3)
- (c) Calculate the total time that it takes the car to stop. (2)
- (d) The car has mass 1100 kg . As the car slows down, it experiences a resistance force that has a constant magnitude of 200 newtons , and a force due to the brakes of magnitude $F \text{ newtons}$.
Find F . (3)

(Total 10 marks)

Q6.

A ball is projected vertically upwards from a height of 5 metres above ground level. Its initial speed is 7 m s^{-1} . Assume that the ball is a particle and that no resistance forces act on it as it moves.

- (a) Show that the maximum height of the ball is 7.5 metres above ground level. (3)
- (b) Show that the time that it takes the ball to reach its maximum height is 0.714 seconds , correct to three significant figures. (2)
- (c) The ball is caught when it is at a height of 2 metres above ground level.
 - (i) Find the total time for which the ball is moving. (4)
 - (ii) Find the speed of the ball when it is caught.

(3)
(Total 12 marks)

Q7.

A car accelerates from rest along a straight road for 5 seconds. At time t seconds, its acceleration, $a \text{ m s}^{-2}$, is given by

$$a = t - \frac{t^2}{5}$$

- (a) By integrating, find an expression for the velocity of the car at time t .

(3)

- (b) Find the velocity of the car at the end of the 5 second period.

(2)

- (c) Find the distance travelled by the car during the 5 second period.

(4)

(Total 9 marks)

Q8.

A cyclist travels along a straight horizontal road. She accelerates uniformly from a speed of 2 m s^{-1} to 6 m s^{-1} as she travels 10 metres.

- (a) (i) Show that the acceleration of the cyclist is 1.6 m s^{-2} .

(2)

- (ii) Find the time that it takes the cyclist to travel this distance.

(2)

- (b) Model the cyclist as a particle of mass 65 kg . A constant resistance force of magnitude 35 N acts on the cyclist. A horizontal force of magnitude F newtons also acts on the cyclist in the direction of motion. Find F .

(3)

(Total 7 marks)

Q9.

A ball is thrown so that it initially travels at 10 m s^{-1} at an angle of 70° above the horizontal.

- (a) A simple model assumes that the ball is thrown from ground level and lands at the same level. Find the time of flight and the range of the ball.

(5)

- (b) A refined model assumes that the ball is at a height of 2 metres when it is thrown, and lands at ground level. Find the range of the ball based on this assumption.

(7)

(Total 12 marks)

Q10.

A cyclist moves from rest along a straight horizontal road. At time t seconds, the displacement of the cyclist from his initial position is s metres.

- (a) For $0 \leq t \leq 10$,

$$s = \frac{t^4}{400} - \frac{t^3}{10} + \frac{3t^2}{2}$$

- (i) Find s when $t = 10$.

(1)

- (ii) Find the velocity of the cyclist when $t = 10$.

(3)

- (iii) Find the acceleration of the cyclist when $t = 10$.

(3)

- (b) For $t \geq 10$ the cyclist moves with a constant velocity, so that

$$s = ht - k$$

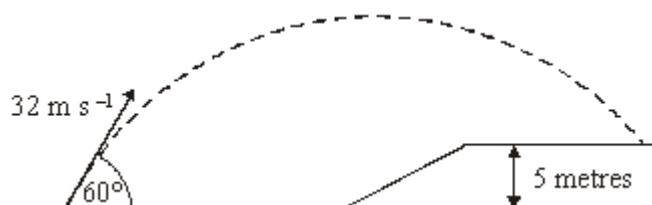
where h and k are constants. Find the values of h and k .

(3)

(Total 10 marks)

Q11.

A golf ball is hit so that it initially travels at 32 m s^{-1} at an angle of 60° above the horizontal. The ball lands on a horizontal surface 5 metres higher than the ground from which it was hit, as shown in the diagram below.



- (a) Show that the ball is in the air for approximately 5.47 seconds.

(5)

- (b) Calculate the horizontal distance travelled by the ball.

(2)

- (c) Find the speed of the ball when it hits the ground.

(5)

(Total 12 marks)

Q12.

A train accelerates uniformly from rest along a straight horizontal track. After it has travelled 400 metres, its speed is 16 m s^{-1} .

- (a) (i) Show that the acceleration of the train is 0.32 m s^{-2} . (2)
- (ii) Find the time that it takes the train to travel the 400 metres. (2)
- (b) When the train has reached a speed of 16 m s^{-1} , its acceleration is increased to 0.5 m s^{-2} .
- (i) Find the distance that the train travels as its speed increases from 16 m s^{-1} to 30 m s^{-1} . (2)
- (ii) Find the total time that the train has been moving when it reaches a speed of 30 m s^{-1} . (3)

(Total 9 marks)

Q13.

A model aeroplane moves in a horizontal plane with a constant acceleration. Initially the aeroplane is at the origin and has velocity $(35\mathbf{i} + 45\mathbf{j}) \text{ m s}^{-1}$. After accelerating for 8 seconds, the velocity of the aeroplane is $(19\mathbf{i} + 13\mathbf{j}) \text{ m s}^{-1}$. The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are perpendicular and lie in the horizontal plane.

- (a) Show that the acceleration of the aeroplane is $(-2\mathbf{i} - 4\mathbf{j}) \text{ m s}^{-2}$. (3)
- (b) Find an expression for the position vector of the aeroplane at time t seconds. (3)
- (c) Find the time when the position vector of the aeroplane is $(300\mathbf{i} + 225\mathbf{j}) \text{ m}$. (7)

(Total 13 marks)

Q14.

A football is kicked from horizontal ground. It initially moves with speed 20 m s^{-1} at an angle of 30° above the horizontal.

- (a) (i) Show that the ball hits the ground approximately 2.04 seconds after it has been kicked. (4)
- (ii) Hence find the range of the ball. (2)

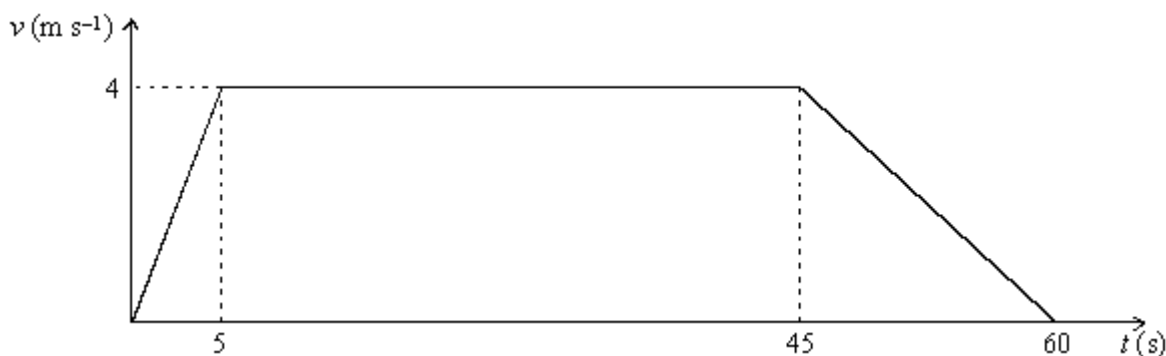
- (b) In fact the ball comes into contact with a player's head when it is at a height of 2 metres. Find the speed of the ball at this height.

(8)

(Total 14 marks)

Q15.

The graph shows how the velocity, $v \text{ m s}^{-1}$, of a cyclist varies with time, t seconds, as she moves along a straight horizontal road.



- (a) Calculate the total distance travelled by the cyclist.
- (b) Find the acceleration of the cyclist during the first 5 seconds of the motion.
- (c) As the cyclist moves, a force of magnitude P newtons acts on the cyclist in the direction of motion. A constant horizontal resistance force of magnitude 50 newtons also acts. Model the cyclist and her bicycle as a particle of mass 65 kg.
- (i) Draw a diagram to show **all** the forces acting on the particle.
- (ii) Find the value of P during the first 5 seconds of the motion.

(3)

(2)

(1)

(3)

(Total 9 marks)

Q16.

A particle moves along a straight line. At time t the displacement of the particle from its initial position is x , where

$$x = 4t + 2e^{-t} - 2$$

- (a) Find the velocity of the particle at time t .
- (b) Find the acceleration of the particle at time t .

(2)

(2)

- (c) Describe what happens to the acceleration of the particle as t increases.

(2)

(Total 6 marks)

Q17.

A particle has mass 2000 kg. A single force, $\mathbf{F} = 1000t\mathbf{i} - 5000t\mathbf{j}$ newtons, acts on the particle, at time t seconds. The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are perpendicular. No other forces act on the particle.

- (a) Find an expression for the acceleration of the particle.

(2)

- (b) At time $t = 0$, the velocity of the particle is $6\mathbf{j} \text{ m s}^{-1}$. Show that at time t the velocity, $\mathbf{v} \text{ m s}^{-1}$, of the particle is given by

$$\mathbf{v} = \frac{t^2}{4}\mathbf{i} + \left(6 - \frac{5t}{2}\right)\mathbf{j}$$

(4)

- (c) The particle is initially at the origin. Find an expression for the position vector, $\mathbf{r}$ metres, of the particle at time t seconds.

(4)

(Total 10 marks)

Q18.

A hot air balloon is released from rest at ground level and rises vertically. It accelerates at a constant rate. It reaches a height of 120 metres after 50 seconds.

- (a) Show that the acceleration of the balloon is 0.096 ms^{-2}

(3)

- (b) Find the speed of the balloon when it is at a height of 200 metres.

(3)

- (c) Find the time that it takes the balloon to reach a height of 200 metres.

(3)

- (d) The mass of the balloon is 250 kg. Find the magnitude of the upward force acting on the balloon.

(3)

(Total 12 marks)

Q19.

A sprinter starts from rest, and accelerates at 2 m s^{-2} for the first 4 seconds of a race. Assume that the sprinter moves along a straight line.

- (a) Find the distance travelled by the sprinter in the first 4 seconds.

(2)

- (b) Find the speed of the sprinter at the end of the first 4 seconds. (2)
- (c) The sprinter then travels at this speed for the remainder of the race. He travels a total distance of 100 metres. Find the total time that he takes to complete the race. (4)
- (Total 8 marks)**

Q20.

A boat moves so that its position vector, $\mathbf{r}$ metres, at time t seconds is

$$\mathbf{r} = 40 \cos\left(\frac{t}{20}\right) \mathbf{i} + 80 \sin\left(\frac{t}{20}\right) \mathbf{j}$$

The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are directed east and north respectively.

- (a) Find an expression for the velocity of the boat at time t . (3)
- (b) In what direction is the boat travelling when $t = 0$? Justify your answer. (2)
- (c) At what time is the boat travelling due south for the first time? (2)
- (Total 7 marks)**

Q21.

A particle moves on a horizontal plane subject to the following three forces:

$$\mathbf{F}_1 = (2\mathbf{i} + 3\mathbf{j})\text{N}, \quad \mathbf{F}_2 = (6\mathbf{i} - 8\mathbf{j})\text{N} \quad \text{and} \quad \mathbf{F}_3 = (-2\mathbf{i} + 10\mathbf{j})\text{N}$$

The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are perpendicular and lie in the horizontal plane.

Do not include any other forces in your calculations.

- (a) Find the resultant force on the particle. (2)
- (b) The mass of the particle is 8 kg. Find the acceleration of the particle. (1)
- (c) At time $t = 0$, the particle is at the origin and has velocity $(3\mathbf{i} - 2\mathbf{j})\text{ms}^{-1}$.
- (i) Find the velocity of the particle when $t = 40$ seconds. (2)
- (ii) Find the distance of the particle from the origin when $t = 40$ seconds. (5)
- (Total 10 marks)**

Q22.

A car accelerates uniformly along a straight horizontal road. As it travels 200 metres, its speed increases from 12 m s^{-1} to 20 m s^{-1} .

- (a) Show that the acceleration of the car is 0.64 m s^{-2} . (3)
- (b) Find the time that it takes the car to travel this distance. (3)
- (c) The mass of the car is 1200 kg. Using the answer from part (a), find the magnitude of the forward force that acts on the car during the above motion if:
- (i) there is not resistance to its motion; (1)
- (ii) a constant resistive force of magnitude 450 newtons acts on the car. (2)

(Total 9 marks)

Q23.

A lift of mass 600 kg is suspended from a single vertical cable. The lift is initially at rest and then starts to accelerate upwards. The tension in the cable remains constant at 6060 newtons while the lift is accelerating. Assume that there is no resistance to the motion of the lift.

- (a) Show that the acceleration of the lift is 0.3 m s^{-2} . (2)
- (b) Find the time that it would take the lift to travel 8 metres. (3)
- (c) Find the speed of the lift when it has travelled 8 metres. (2)

(Total 7 marks)

Q24.

At time $t = 0$, a particle is at the origin and moving with velocity $(4\mathbf{i} + 2\mathbf{j}) \text{ m s}^{-1}$. When $t = 10$ seconds the position vector of the particle is $(44\mathbf{i} + 28\mathbf{j})$ metres. The particle is moving with constant acceleration.

- (a) Find the acceleration of the particle. (4)
- (b) Find the position vector of the particle at time t . (3)
- (c) At time $t = T$, the position vector of the particle is $(96\mathbf{i} + 72\mathbf{j})$ metres.
- (i) Find T .

(4)

(ii) Find the speed of the particle at this time.

(3)

(Total 14 marks)



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